

## JACKSONVILLE CRAIG, FL, USA

WMO: 747820

Lat: 30.336N

Lon: 81.515W

Elev: 41

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 53860

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 32.9       | 36.3 | 16.6                          | 12.7 | 41.5 | 21.2 | 15.9 | 43.5 | 22.6                  | 58.3 | 20.4 | 55.1 | 5.6                   | 340  | 0.383 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.3                   | 94.4            | 76.8 | 92.3 | 76.6 | 90.5 | 76.4 | 80.0                | 88.1 | 79.2 | 87.3 | 78.4 | 86.2 | 8.8                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.2                            | 146.7 | 82.9 | 77.1 | 141.3 | 82.4 | 76.2 | 137.1 | 82.0 | 43.3          | 87.9 | 42.5 | 87.2 | 41.8 | 86.6 | 86.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 19.2              | 17.5 | 15.9 | DB                         | 27.1 | 97.5               | 2.9 | 2.0                                                | 25.0 | 98.9       | 23.3 | 100.0      | 21.7 | 101.1      | 19.6 | 102.6 |
|                   |      |      | WB                         | 24.0 | 81.6               | 3.1 | 1.0                                                | 21.7 | 82.3       | 19.9 | 82.9       | 18.2 | 83.5       | 16.0 | 84.2  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 69.8   | 54.8  | 58.1  | 62.5  | 68.7  | 75.0  | 80.4  | 82.2  | 81.9  | 79.1  | 72.4  | 63.6  | 58.2  |
|                                                                        | DBStd       | 11.55  | 9.07  | 8.61  | 7.57  | 6.03  | 4.69  | 3.38  | 2.68  | 2.57  | 3.18  | 6.11  | 7.52  | 8.88  |
|                                                                        | HDD50       | 117    | 57    | 23    | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 26    |
|                                                                        | HDD65       | 1097   | 333   | 220   | 137   | 33    | 1     | 0     | 0     | 0     | 0     | 15    | 114   | 244   |
|                                                                        | CDD50       | 7343   | 204   | 251   | 393   | 562   | 776   | 911   | 998   | 872   | 694   | 411   | 282   |       |
|                                                                        | CDD65       | 2848   | 16    | 27    | 58    | 145   | 312   | 461   | 533   | 524   | 422   | 245   | 71    | 34    |
|                                                                        | CDH74       | 25214  | 69    | 154   | 381   | 1031  | 2616  | 4329  | 5524  | 5299  | 3690  | 1646  | 332   | 143   |
|                                                                        | CDH80       | 8896   | 3     | 14    | 60    | 239   | 875   | 1712  | 2344  | 2082  | 1173  | 347   | 38    | 9     |
| (14) Wind                                                              | WSAvg       | 6.9    | 7.4   | 7.5   | 7.9   | 7.7   | 7.2   | 6.4   | 6.0   | 5.8   | 6.5   | 6.6   | 6.7   | 6.9   |
| (15) Precipitation                                                     | PrecAvg     | 50.3   | 3.1   | 2.7   | 3.3   | 2.6   | 2.9   | 6.8   | 6.1   | 6.6   | 7.2   | 4.5   | 2.0   | 2.5   |
|                                                                        | PrecMax     | 72.2   | 8.1   | 9.4   | 8.9   | 7.4   | 12.5  | 20.3  | 14.0  | 15.4  | 14.0  | 10.9  | 4.8   | 8.7   |
|                                                                        | PrecMin     | 29.2   | 0.3   | 0.6   | 0.4   | 0.6   | 0.5   | 2.0   | 2.0   | 2.4   | 2.9   | 0.2   | 0.0   | 0.4   |
|                                                                        | PrecStd     | 10.2   | 2.0   | 1.9   | 2.0   | 1.7   | 2.6   | 3.8   | 2.5   | 3.2   | 3.2   | 3.2   | 1.3   | 2.0   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 80.1  | 82.1  | 84.9  | 89.2  | 93.4  | 97.3  | 97.1  | 96.3  | 92.7  | 88.5  | 83.8  | 81.4  |
|                                                                        |             | MCWB   | 67.8  | 68.0  | 67.8  | 69.9  | 72.6  | 76.8  | 77.3  | 77.4  | 75.5  | 75.3  | 72.4  | 71.0  |
|                                                                        | 2%          | DB     | 75.5  | 79.0  | 81.4  | 85.7  | 90.6  | 93.6  | 93.8  | 93.1  | 90.1  | 85.7  | 80.8  | 78.4  |
|                                                                        |             | MCWB   | 66.1  | 67.0  | 67.4  | 69.5  | 72.1  | 76.4  | 77.5  | 78.0  | 75.4  | 74.4  | 70.4  | 69.2  |
|                                                                        | 5%          | DB     | 72.4  | 75.0  | 78.7  | 82.4  | 87.9  | 90.9  | 91.6  | 90.8  | 88.1  | 83.7  | 78.2  | 74.7  |
|                                                                        |             | MCWB   | 64.7  | 65.1  | 65.7  | 68.6  | 72.1  | 76.2  | 77.2  | 77.8  | 75.6  | 73.4  | 68.9  | 67.8  |
|                                                                        | 10%         | DB     | 69.5  | 71.8  | 75.1  | 80.2  | 84.7  | 88.4  | 90.1  | 89.2  | 86.1  | 81.7  | 75.0  | 71.7  |
|                                                                        |             | MCWB   | 63.1  | 63.8  | 64.1  | 67.4  | 71.7  | 76.0  | 77.4  | 77.7  | 75.4  | 72.3  | 67.4  | 66.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 70.4  | 70.9  | 72.4  | 74.2  | 77.1  | 80.3  | 81.2  | 81.2  | 79.5  | 78.3  | 75.4  | 72.9  |
|                                                                        |             | MCDB   | 75.2  | 77.4  | 79.5  | 82.5  | 85.3  | 89.7  | 88.6  | 88.8  | 85.4  | 83.8  | 80.6  | 77.9  |
|                                                                        | 2%          | WB     | 68.4  | 69.1  | 69.6  | 72.6  | 75.6  | 78.8  | 79.9  | 80.1  | 78.3  | 77.0  | 73.1  | 71.0  |
|                                                                        |             | MCDB   | 73.6  | 75.5  | 77.5  | 80.4  | 83.6  | 88.2  | 88.2  | 88.0  | 84.7  | 82.4  | 77.4  | 75.8  |
|                                                                        | 5%          | WB     | 66.3  | 67.3  | 68.1  | 71.2  | 74.7  | 77.9  | 79.0  | 79.2  | 77.7  | 75.8  | 71.2  | 69.1  |
|                                                                        |             | MCDB   | 71.2  | 73.2  | 75.1  | 78.7  | 82.5  | 86.9  | 87.5  | 87.1  | 84.2  | 81.1  | 75.3  | 73.4  |
|                                                                        | 10%         | WB     | 64.0  | 65.1  | 66.4  | 69.9  | 73.5  | 77.0  | 78.2  | 78.4  | 77.0  | 74.7  | 69.3  | 66.9  |
|                                                                        |             | MCDB   | 68.0  | 70.3  | 72.7  | 77.3  | 81.2  | 85.5  | 86.6  | 85.9  | 83.4  | 79.8  | 73.9  | 70.9  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 19.4  | 19.3  | 19.6  | 19.5  | 18.6  | 16.5  | 16.3  | 14.9  | 13.6  | 15.7  | 18.2  | 18.0  |
|                                                                        |             | MCDBR  | 21.4  | 21.7  | 22.7  | 21.6  | 21.6  | 19.5  | 18.1  | 17.2  | 16.6  | 16.3  | 18.8  | 18.8  |
|                                                                        |             | MCWBR  | 13.2  | 11.7  | 11.0  | 9.5   | 8.0   | 6.9   | 6.4   | 6.2   | 6.4   | 7.4   | 10.1  | 11.2  |
|                                                                        | 5% WB       | MCDBR  | 18.5  | 18.7  | 19.2  | 18.9  | 18.0  | 17.4  | 16.5  | 15.4  | 13.7  | 12.9  | 15.0  | 16.5  |
|                                                                        |             | MCWBR  | 12.5  | 11.2  | 10.7  | 9.6   | 7.8   | 6.9   | 6.9   | 6.5   | 6.0   | 6.4   | 9.5   | 10.7  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.338 | 0.349 | 0.372 | 0.396 | 0.437 | 0.480 | 0.505 | 0.489 | 0.428 | 0.401 | 0.361 | 0.346 |
|                                                                        | taud        |        | 2.534 | 2.508 | 2.455 | 2.373 | 2.298 | 2.238 | 2.173 | 2.221 | 2.391 | 2.430 | 2.493 | 2.536 |
|                                                                        | Ebn at Noon |        | 284   | 291   | 290   | 285   | 272   | 259   | 252   | 255   | 268   | 269   | 273   | 275   |
|                                                                        | Edh at Noon |        | 28    | 31    | 35    | 39    | 42    | 45    | 48    | 45    | 36    | 33    | 29    | 26    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 987   | 1183  | 1546  | 1892  | 2027  | 1890  | 1880  | 1751  | 1505  | 1327  | 1073  | 889   |
|                                                                        | RadStd      |        | 70    | 133   | 142   | 116   | 121   | 125   | 121   | 111   | 125   | 114   | 61    | 76    |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            |       | +0.97   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +308  | +239  |
| (47) Regional (33 neighbors) |       | +0.77   | N/A    | N/A     | N/A   | N/A   | N/A         | -20   | -115  | +288  |

Nomenclature: See separate page